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Thirty-five years after *Farmer First:* embedding farmer participation in national agricultural policy and practice

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Abstract
Thirty-five years ago, the *Farmer First* movement presented a new paradigm for agricultural R&D, making the case for a more prominent role for farmers in the design, development, and dissemination of agricultural technologies and practices. Since then, a diverse range of projects have been implemented with explicit attention to the engagement and participation of farmers and farmer-based organizations. After nearly four decades, it is worth reflecting further on the full extent of this paradigm's impact on research and innovation policy. Specifically, has the movement led to the institutionalization of meaningful farmer engagement and participation in national policy formulation and implementation? To answer this question, we adopt a case study approach and combine project and public policy document analysis with key informant interviews held with actors in government, research institutions, donor agencies, and farmer organizations. We select three farmer-first-oriented projects in Honduras, Mali, and the eastern Africa region, each of which have had long exposure to the participatory principles we expect to see reflected in the agricultural policies of the country. We document the specific interactions between project actors and the policy space. We then analyze common policy trends across case studies. First, we find that change is often driven by highly motivated individuals who operate with a lack of buy-in from policy-relevant institutions. Second, we observe change when organizations practicing farmer-first approaches partner with more influential organizations, which may not have internalized those principles but have a common interest in the dissemination of technologies and practices. Third, we find that national agricultural policies in the case study countries are not codifying well-established participatory practices. We discuss pathways toward greater farmer participation in light of these findings.

Keywords Farmer first · Participatory research · Honduras · Mali · Eastern Africa · Agricultural policy

Introduction

Thirty-five years ago, scholars and practitioners in the field of agriculture argued that agricultural research and development in the post-Green Revolution era required a more influential role for farmers and farmer organizations in the innovation process. These ideas spawned thousands of participatory initiatives that came to be known collectively

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as the *Farmer First* movement. The movement was documented in a vast corpus of literature that provided conceptual frameworks, documented practices, case studies, and a considerable amount of reflection and analysis (Chambers et al. 1989; Farrington 1989; Scoones and Thompson 1994, 2009; Thompson and Scoones 1994). The movement also led to the development of a diversity of participatory research and extension methods that were tested and applied in low- and middle-income countries, often by public research and non-governmental organizations, and often in projects concerned withcrop improvement, plant breeding, and seed systems. The *Farmer First* approach advocated for a collaborative model where farmers and researchers work together to analyze problems and co-design, implement and assess context-specific solutions.

The conceptual origins of *Farmer First* include several strands of work on how to conduct research with farmers that can be traced back to the 1970s and 1980s, all of which have sought to democratize knowledge production and innovation in agriculture. These include the Sondeo approach in Latin America (Hildebrand 1981), farming systems research (Amanor 2024; Darnhofer et al. 2012; Collinson 2000; Biggs and Clay 1981), agroecosystem analysis (Conway 1985) and participatory rural appraisal (Lovelace, et al. 1987; Chambers 1994), among others. *Farmer First* also grew out of frustrations with the "Transfer-of-Technology" model, which was seen as favoring resource-rich farmers at the expense of resource-constrained small-scale farmers (Chambers and Jiggins 1987a, 1987b).

Farmer First proponents called for a paradigm shift in agricultural research (Chambers and Ghildyal 1985). This shift entailed focusing on the farm family and the farming system in which they operate, prioritizing holistic and interdisciplinary appraisals of farm families' resources and requirements, and emphasizing R&D that is both conducted on the farm and with the farmer, with scientists having a more consultative and collegial role. The principles, concepts and methods underpinning this shift embraced diverse forms of knowledge, co-design and co-management of innovation processes, people-centred learning, and action research in agriculture.

In plant breeding and seed systems research, visions of what a farmer-first approach looks like has changed over time. They have varied from jointly developing new crop varieties using participatory breeding and varietal selection (see Ceccarelli and Grando 2020 for a review) to promoting farmer field schools that offer high-intensity experiential learning opportunities around the cultivation of new varieties (Waddington et al. 2014; Davis 2006), to advancing digitally mediated information crowdsourcing that captures input on varietal attributes from thousands of farmers simultaneously (van Etten et al. 2019; de Sousa et al. 2024). Researchers and practitioners also expanded their definition of who a farmer is to recognize heterogeneity in farming populations. This includes not only women and men smallholder farmers, but also youth farmers, landless laborers, livestock keepers, fisher people, and peri-urban farmers, each characterized by different and ever-changing assets, capabilities, voice and agency, and each operating within distinct and complex agroecological conditions, farming systems, and livelihood strategies (Ruben and Pender 2004; K et al. 2016; Herrero et al. 2014; Huber et al. 2024). Across this diversity, one widely shared expectation has persisted: that organizations responsible for agricultural R&D should evolve to encourage farmers' voice, agency, and power in deciding priorities, generating evidence, and promoting policies for inclusive agricultural innovation (Ashby 2013; James and Sulemana 2014).

Some countries have formulated novel agricultural policies and programs that meet this expectation, and have actively involved farmers in the conduct of agricultural R&D. For example, the Indian Council of Agricultureuses participatory approaches to research and extension through its *Krishi Vigyan Kendras* or "Farm Science Centers" (Sahoo et al. 2021; Joshi et al. 2012; Witcombe and Yadavendra 2014). EMBRAPA in Brazil employs farmer participatory approaches to develop and disseminate agroecological farming practices (Ekboir 2003; Alves 2012; Nehring 2024). Public agricultural research organizations in Nepal, Bangladesh, the Philippines, Columbia, Peru, Bolivia, and Ecuador (Thompson 1995) and public projects in Uganda (Mwanga et al. 2011), Kenya (Lilja and Bellon 2006), and Ethiopia (Erkossa et al. 2022) have all incorporated elements of a participatory approach. Increasingly, organizations and firms at more downstream points of agricultural value chains – processors, retailers, financial service providers – have similarly introduced participatory approaches to their work with farmers (Thiele et al. 2001; Reardon et al. 2009, 2019; Chambers et al. 2023).

Yet despite these signs of progress, there is still relatively limited discussion of the extent to which farmers are being engaged in the research agendas and policy processes that shape their roles within increasingly market-driven food systems (Occelli et al. 2024). A review of more than six million research articles in the past 14 years found that less than 10% of scientific research focuses on farmers and related food system actors (Porciello et al. 2024). This near absence of work that meaningfully engages farmers, and other actors may hamper the institutionalization and scaling of participatory approaches that generate the insights and evidence needed for inclusive national agricultural policies and practices (Ceccarelli and Grando 2019).

This leads us to ask: *After nearly four decades of accumulated evidence and experience in participatory approaches,*

have national agricultural policies (i.e., their design process and implementation practices) institutionalized meaningful farmer participation, incorporating the Farmer First principles of including farmers as co-researchers and joint decision-makers?

To answer this question, we adopt a case study approach and combine the analysis of project and public policy documents with key informant interviews held with actors from government, research institutions, donor agencies, and farmer organizations. We select three farmer-first-oriented projects in Honduras, Mali, and the eastern Africa region, whose long exposure to participatory principles we expect to see reflected in agricultural policies of the country. We characterize the specific interactions between each project and the policy space, alongside a synthesis of common patterns across the cases. We focus on drawing out shared lessons and insights without necessarily seeking to identify clear causal connections.

Since Ashby and Sperling (1995) and Thompson (1995) reviewed ways in which institutions have responded to key characteristics of participatory R&D in agriculture, numerous studies have discussed national agricultural policies in light of participatory R&D. Examples include Sperling and Ashby (2001) on typologies of participation in agricultural planning, Hall et al. (2003) on innovation systems and institutional learning and, Dinesh et al. (2021) on institutionalizing knowledge generation-policy engagement in agricultural research and development. The body of literature focusing on the design, governance, and management of participatory multistakeholder platforms for agricultural innovation is also vast (e.g., Brouwer et al. 2013; Herens et al. 2022; Ratner et al. 2022). For example, Akpo et al. (2015) focus on the relevance of process design in shaping engagement and relations within the platforms, while Vallejo and Hauselmann (2004) review the proliferation of these platforms and their legitimacy in political discourse. Thorpe et al. (2022), Pamuk and Van Rijn (2019), and Pamuk et al. (2015, 2014) collect evidence of the impact and effectiveness of these platforms and provide generally supportive but highly nuanced results.

Our study enriches this discussion by centering on four farmer-first principles of policy change, inspired by the reflections in Ashby (2022). Building on these four principles, this study is, to the best of our knowledge, the first to explore whether and how national agricultural policies have institutionalized meaningful farmer participation. In a world that is characterized by considerable risk and uncertainty attributable to anthropogenic climate change, global economic interdependence, and geopolitical factors, it is worth asking whether farmer participation in agricultural R&D and national agricultural policy remains as much a concern today as it was 35 years ago.

The remainder of the paper is organized as follows. Section "The Farmer first principles applied to policy change" presents the theoretical framework that shapes our investigation, section "Data and methods" explains the data and methods used for the case study analysis, while section "Case study description" introduces the reader to the three case studies. Results presented in section "Results". Section "Discussion" discusses the implications of our findings for food policies and practices in low- and middle-income countries.

The *Farmer first* principles applied to policy change

Ashby (2022) synthesizes broad dimensions of a political agenda which can guide the embedding, operationalizing, and scaling of farmer-centered practices into national agricultural policies and practices. We call these dimensions, farmer-first principles of policy change. They are:

1 Inclusive Governance: Institutionalize farmers and farmer organizations as co-decision-makers in policy design, implementation, monitoring, and evaluation. This moves beyond tokenistic consultation toward shared authority.
2 Responsive Innovation Systems: Strengthen demand-side accountability by creating mechanisms that prioritize farmer-defined needs and capabilities, not merely market signals.
3 Decentralized Knowledge and Power: Adapt research and extension systems to local agroecological and sociocultural contexts, enabling farmers to co-create solutions and influence resource allocation.
4 Transformative Education: Embed participatory research principles in agricultural curricula to cultivate professionals who view farmers as partners rather than passive beneficiaries.

These dimensions act as common principles, intended as normative and analytical anchors, whose concrete expression is culturally and contextually contingent. What inclusive governance, agency, or demand looks like in practice is shaped by the historical, political, and cultural setting in which farming takes place. It follows that integration of farmer-first principles into policy need not, and arguably should not, look identical contexts.

Nonetheless, building on the synthesis of Ashby (2022), we suggest that these four principles, when embodied in long-term programs, projects and activities, can contribute over the years to discernable policy changes that are aligned with farmer-first principles. Specifically, we hypothesize that four conditions can lead to national agricultural policies

and practices that are more farmer-led: intentionality in engaging farmers and their representative organizations in shaping food policy processes (*the politics of policy*); the capacity of national agricultural institutions to respond to context-specific and farmer-specific needs and preferences (*the politics of demand*); the decentralization of power, production and knowledge to accommodate the changing role of farmers in driving innovation processes (*the politics of knowledge*); and a system of higher education institutions actively training the next generation of agricultural researchers, practitioners and policymakers capable of working constructively *with*, rather than *on* farmers (*the politics of andragogy*).

Farmer-first principles of policy change intersect with donor influence. This is because national agricultural policies in many low- and middle-income countries are developed in a context that is influenced by multilateral, bilateral, and philanthropic donors, whose funding and investment strategies steer national priorities (Birner and Resnick 2010; Mockshell and Birner 2015) or challenge them (New York Times 2007).

Global donor strategies often privilege a more "functional-instrumental participation" (Jackson-Smith and Veisi 2023) – accelerating technology adoption – while marginalizing normative empowerment. This conflation of "demand" with "farmer" reduces agency, narrowly positioning farmers as market actors rather than rights-bearing citizens with their own development priorities and capabilities. Other donors, though smaller in scale, tend to support empowerment-oriented participation, fostering social capital and farmer-led innovation. Recognizing these dynamics is critical as policy change is not merely technical but deeply political, shaped by competing narratives of efficiency, equity, and agency.

Our framework thus situates farmer-first principles within broader governance and funding architectures, emphasizing that sustainable policy transformation depends on aligning national priorities with ethical commitments to voice, justice, and epistemic diversity.

Data and methods

To answer our research question, we opted for a case study approach, which is a rigorous, detailed, spatially bounded, temporally limited description and analysis of an event, intended by the researcher to exemplify a more general theory (Creswell and Miller 1997). For us, a case study is a project that is known to have embodied farmer-first principles. Case studies are increasingly used in food and agricultural policy (e.g., Naylor et al. 2023).

In studying if and how national policies have institutionalized meaningful farmer participation, we hypothesize the interplay of a series of interconnected interventions, resulting in crucial learning moments and adaptations over an extended period and involving sustained interactions by a diverse group of local, national and international actors. We based our case study approach on the CIAMO (Context-Intervention-Actor-Mechanism = Outcome) framework[1] (Greenhalgh et al. 2004; Mukumbang 2021; Mukumbang et al. 2018, 2020), to reflect these "learning loops" (Figure A1 of the Supplementary Materials). Through the CIAMO framework, we examined the context, interventions, actors, mechanisms, and intermediate and longer-term outcomes of a set of case studies against the four farmer-first principles of policy change (Ashby 2022), described in section "The Farmer first principles applied to policy change". The analysis is corroborated by extensive policy textual analysis and key informant interviews. Figure 1 summarizes the main steps taken for this study.

We first conducted a review of the three *Farmer First* books, along with publicly available workshop summary and reflection briefs related to farmer-first principles. We then developed a working definition for "national agricultural policy", which highlights aspects of their design process and implementation practices. National agricultural policies include but are not limited to laws, rules, directives, guidelines, practices, and processes, made by governments that affect the production, distribution, and consumption of food both in a country and between or among countries.

Accordingly, we consider national agricultural policies and practices to include codified legislation such as food and agriculture strategies, acts, policies, regulations, investment and expenditure plans, and specific programs at a national, subnational, agroecological, commodity, or market level. We also consider the term to include practices, habits, and routines that are not codified but which have permeated *de facto* into food and agricultural organizations and institutions. For example, research protocols routinely employed by a research center's scientists, or decentralized routines in the provision of extension services or seeds. Our definition is consistent with prior descriptions of the term, e.g., Pinstrup-Andersen and Watson (2011) or Maxwell and Slater (2003), albeit less detailed.

Case study selection

We proceeded by selecting case studies to analyze. A priori, we were looking for cases which attempted to change

[1] These configurations are sometimes referred to as 'ICAMO' in the literature, placing the intervention before the context (e.g., De Weger et al. 2020; Mukumbang et al. 2018), but we prefer 'CIAMO', to emphasize the point that the context exists before an intervention is implemented, which then catalyzes a series of change processes within that particular socio-technical system.

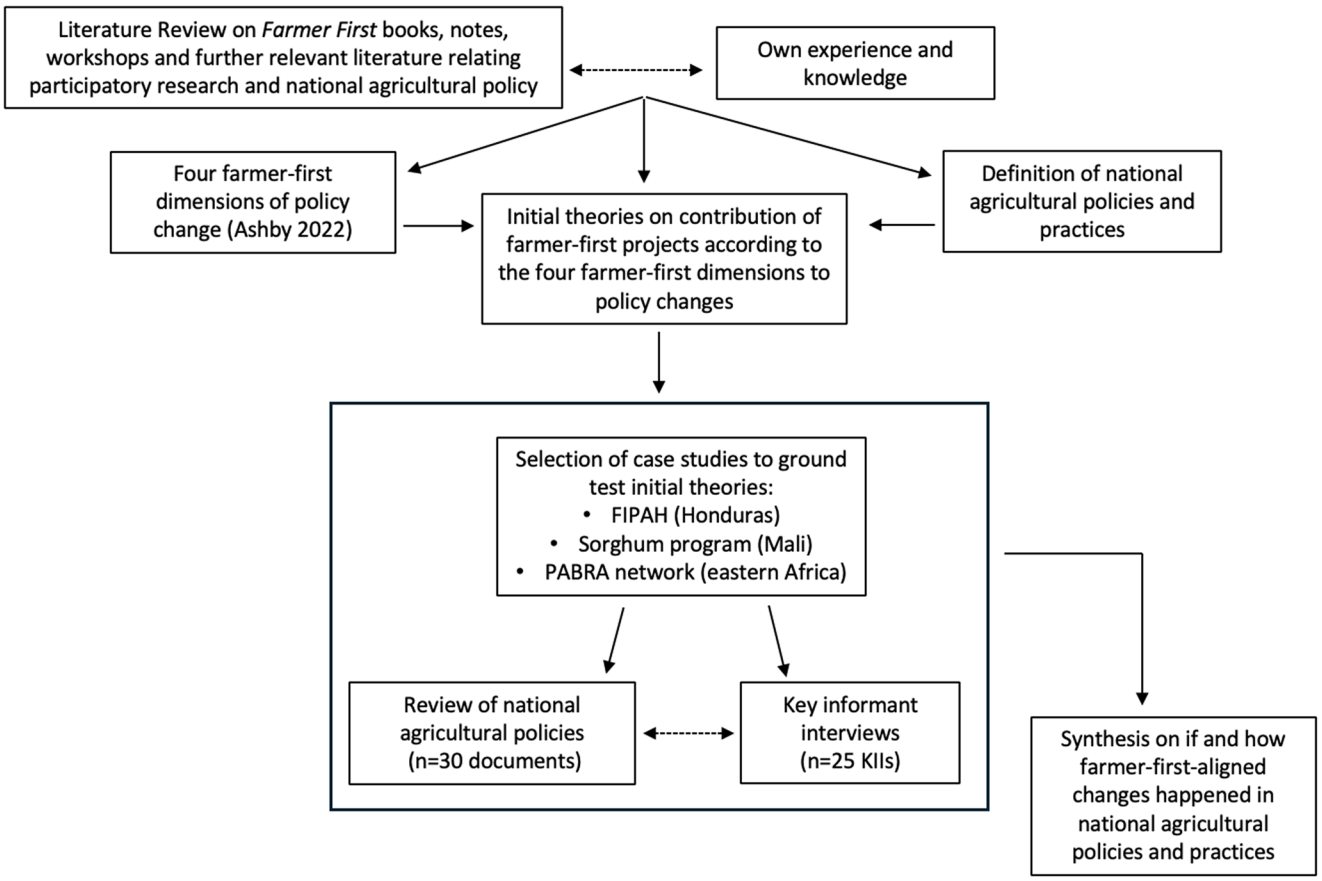


Fig. 1 Steps taken in this study. Dashed lines indicate recursive steps

national agricultural policy, so that we could explore how this process developed. The selection criteria were that the project: (i) pertained to the domain of plant breeding and/or seed system development; (ii) brought geographical diversity; (ii) showed a significant and sustained degree of exposure to the farmer-first principles; (iv) was implemented at a national or regional scale and across a period of time of at least a decade, to allow for the production of a trail of peer-reviewed papers, interactions with policy change processes in their focus countries, and a critical mass of key informants.

Following these criteria, the FIPAH project in Honduras (FIPAH Honduras, hereafter), the sorghum program in Mali (Sorghum Mali, hereafter), and the PABRA network in eastern Africa (PABRA eastern Africa, hereafter) were selected as case studies. Each program has at least two decades of continuous engagement with farmers: FIPAH Honduras has been active since the early 2000s; the Mali sorghum program's participatory work dates back to 1998; and PABRA eastern Africa has operated since 1996. Each pertained to the domain of plant breeding and seed systems: FIPAH Honduras and PABRA eastern Africa on beans; Mali on sorghum. Each case was guided by farmer-first principles and recognized as associated with the *Farmer First* movement: in all three cases, lead researchers were invited to the *Farmer First* workshop and/or featured in the second *Farmer First* book (Scoones and Thompson 2009) in early 2000s. Finally, each case study engaged extensively with policy actors in the country, including national agricultural research organizations and ministries of agriculture. And each was supported by a rich body of published peer-reviewed evidence.

The three cases span distinct policy architectures, agro-ecological systems, and sociocultural settings. The fact that they cover a variety of crops, locations, implementation scale, donor types and stages of policy institutionalization allows us to synthesize not only if, but most importantly how, participatory farmer-first principles have been put into practice in different contexts in the past 35 years.

Following the case study selection, two experts for each case study were invited to co-author this study as a means of

ensuring that context-, project-, and policy-specific dynamics were adequately captured and represented.[2]

Document evidence and key informant interviews

For each case study, we collected evidence through analysis of national agricultural policies and key informant interviews. First, we conducted a policy review which included relevant national policies in the domain of crop production, seed systems, crop marketing and distribution (internal market and export), and policies reflecting the agricultural strategy of the country more broadly. When available, teaching curricula and national teaching policies on agriculture were collected. Agricultural policies developed in response to international agreements (e.g. regional strategies) were also included. A total of 30 policy documents were reviewed, in English, Spanish (for FIPAH Honduras case) and French (for Sorghum Mali case) (full list available in Annex 2 of the Supplementary Materials).

The list of relevant policies within each case study was iteratively revised, with policy documents being identified through desk reviews and during key informant interviews. A policy review protocol was used to extract evidence along the four farmer-first principles of policy change (Annex 1 of the Supplementary Materials).

Key informant interviews were conducted between January and April 2025. Three categories of respondents were interviewed: project actors (e.g., project leadership, research and implementation team engaged, donors, farmers' organizations); national policy actors (e.g., officers of national agricultural research organizations and other government entities at the local and national levels); and professors, lecturers, researchers, and other scholars engaged in national agricultural universities and colleges. Informed consent was obtained from all informants. The interview protocol was shared and reviewed by the Institutional Review Board for Human Participants at Cornell University, which confirmed that this research did not require IRB oversight.[3] Respondents were interviewed following a semi-structured protocol based on the CIAMO framework, with the goal of collecting evidence on the contribution of farmer-first principles to policy change. Questions were grounded on the synthesis of Ashby (2022) and were designed to: differentiate engagement led by researchers and experts from farmer-led work in the specific case study (Principle 1: inclusive governance), to investigate types of demand-led responsiveness for the agricultural innovation at the center of the case study (Principle 2: responsive innovation systems), to identify if/how location-specific farmers' priorities (both geographical and cultural) were considered in policies (Principle 3: decentralized knowledge and power) and to assess whether changes to teaching curricula could be observed as an outcome of the case study (Principle 4: transformative education). For each respondent, we also collected the gender, the role, the history and years of engagement in the project. The full protocol is available in Annex 3 of the Supplementary Materials.

With respect to policy documents, key informant interviews provided an "on-the-ground" perspective of what worked and what did not work in the institutionalization of farmer-first principles.

A total of 25 interviews were conducted across the three case studies (5 for FIPAH Honduras, 13 for Sorghum Mali and 7 for PABRA eastern Africa). Informants for FIPAH Honduras included project leadership and staff, a bean-breeding researcher, and a donor representative, selected for their direct involvement in the program's engagement with national seed and breeding policy. For Sorghum Mali, informants spanned breeders, farmer-organization leaders and members, university lecturers, and seed-system actors, reflecting the case's emphasis on farmer cooperatives. Finally, for PABRA eastern Africa, interviews included PABRA leadership, affiliated researchers, a market researcher, and national research-system staff in member countries, selected to capture both the regional and the national levels of PABRA's engagement.

Interviews were conducted online, in English and French (for the Sorghum Mali case), recorded, and transcribed.

Analysis

A deductive (framework-guided) thematic analysis against the CIAMO categories and the four farmer-first principles of policy change was conducted for each case study individually. Quotes from documents and interviews were analyzed and synthetized case by case. Context, key actors and intervention of each case study are introduced in section "Case study description", while we summarize findings on mechanisms and outcomes of each case study in section "Results".

Subsequently, comparing across case studies, we took an inductive step across the coded material to identify cross-cutting patterns which identify differences and similarities between case studies. This allowed us to abstract general patterns of if/how farmer-first principle of policy change contributed to the meaningful institutionalization of

[2] The specific roles of each co-author in this study is provided at the end of the paper. All co-authors who were directly involved in implementation of the project that is featured as a case study were not directly involved in designing the study design, presenting the results, or drawing the conclusions. However, they did play a critical role in providing essential data on project andpolicytopics,andinvalidatingt heanalysis,results,andconclusionspresentedhere.Theirinclusionasco-authorsisconsistent with the guidelines of the institutions with which the main authors are affiliated.

[3] This decision tree guideline was followed: https://researchservices.cornell.edu/sites/default/files/2019-05/IRB%20Decision%20Tree.pdf.

farmers' participation into agricultural R&D, across crops, geographies and institutional contexts. We discuss what these lessons learnt imply in section "Discussion".

Case study description

In this section, we present the context, intervention and key actors characterizing each case study.

Case study 1: FIPAH program in Honduras

Context. Smallholder farmers in hillside areas of Honduras are vulnerable to seasonal food scarcities, aggravated by isolation, narrow markets and climate change-induced reductions in grain yields and outputs (Gomez et al. 2025). National agricultural policies have directed investment towards a centralized breeding program that has produced maize and bean varieties that typically fail to account for agroecological and socioeconomic heterogeneity, particularly in marginal areas where small farmers are located (Gomez et al. 2025; Dodd et al. 2023). It follows that varieties developed by these breeding programs failed to deliver yield-enhancing varieties to hillside communities. This led some breeders in the national agricultural research system to shift towards participatory plant breeding and varietal selection of maize and beans varieties with the goal of adapting materials to more localized conditions (Humphries et al. 2005, 2008, 2015). Their work gave rise to La *Fundación para la Investigación Participativa con Agricultores de Honduras* (FIPAH),[4] which has been operating since the early 2000s. The Foundation was launched by a group of Honduran agronomists working since 1993 in the participatory research program of the International Centre for Tropical Agriculture (CIAT), called IPCA (*Investigación Participativa en Centro América or Participatory Research in Central America*). Adopting from IPCA a community-based agricultural intervention called local farmer research committees (CIAL, in the Spanish acronym) (Ashby et al. 1995, 2000), FIPAH has worked in rural Honduras for the past 25 years with long-term support from multiple donors.

Intervention. In the hillside regions of Honduras, agronomists organized local farmer research committees (CIALs) (Gomez et al. 2025; Dodd et al. 2023). The process is described in detail in (Gomez et al. 2025; Dodd et al. 2023). FIPAH used the CIAL methodology to engage poor, hillside farmers in collaborative research, farmer field schools and participatory varietal selection. As a result, FIPAH-supported CIALs have contributed to 12 new bean varieties and 7 maize varieties since 2005. A key feature of this process intervention is that FIPAH comprised change agents, who championed farmer engagement in long-term policy dialogue and institution-building at national and local levels.

Key actors. This project was led by FIPAH, in collaboration with key actors in the Zamorano University, CIAT, and the University of Guelph. FIPAH has been supported since 2000 to the present, by Seed Change, a non-governmental organization formerly known as USC Canada, with funding from the Canadian Government and, between 1995 and 2000, from the International Development Research Centre (IDRC).

Case study 2: Sorghum program in Mali

Context. Sorghum is cultivated across Mali in a wide range of ecologies from the Sahel to the Guinea Savannah zone, all with highly variable rainfall and degraded soils, agricultural development policy experienced a surge during the mid-late 1990s, when a democratically elected government opened opportunities for farmer organizations to influence the cotton sector (Bingen 1998) and deregulated cereal markets (Dioné 2000). Nonetheless, as the agricultural sector kept experiencing low adoption of improved varieties and low productivity gains, the role of agricultural research and its operating model for achieving benefits for farmers were questioned. This was particularly evident for the sorghum breeding program (Yapi et al. 2000). This context created the opportunity for introducing participatory approaches in the program. Parallelly, legislation and regulations governing variety release, seed production and commercialization changed significantly during the same period. Seed legislation introduced in 1995 defined actors and processes very generally and stipulated that only certified seed could be sold. The release of the 2010 seed policy and law[5], followed by new decrees and orders (2017, 2018, 2019) regulate the variety release processes in detail, specifically in view of facilitating private investments. In this context, farmer cooperatives – historically strong in the country- are not considered enterprises but are allowed to sell certified seed.

Intervention. Sorghum breeders with participatory breeding experiences (Weltzien et al. 2006) were transferred to Mali to reorient ICRISAT's West African Sorghum breeding program. The team initiated farmer participatory evaluations of germplasm and trained sorghum breeders on participatory approaches (Weltzien et al. 2006). After the initial learning phase, the program assisted two farmer organizations towards financial stability, governance, technical efficiency for coordinating farmer managed variety trials, seed

[4] In English, this translates to "Foundation for Participatory Research with Farmers of Honduras".

[5] The Malian seed laws were released in 2010 to comply with the ECOWAS (Economic Community of West African States) regional seed laws released in 2008 complying to the WTO regulations, and thus UPOV1991 protocols.

production and dissemination. Critical for breeding suitable varieties was the inclusion of thresholds for grain quality using the concept of "food yield" (Isaacs et al. 2023) and the focus on local adaptation, for example the importance of adaptation to low soil phosphorus availability (Leiser et al. 2018). Many farmers and their organizations participated in exchange and learning workshops, technical training programs, training on radio communication, cooperative governance and basic business skills, field days and discussions at research stations, and exchange visits in other regions and neighboring countries. All variety releases were made with names of collaborating farmer organizations and scientists involved (Siart 2008).

Key Actors. The Mali sorghum team had breeders from Institut d'Economie Rurale (IER) and ICRISAT, and Institut Polytechnique Rural (IPR). IER engaged farmers as a part of Committees of Users of Research Results (CRU). Extension services and NGOs (ACOD, World Vision, and SNV) facilitated initial farmer engagement. The primary farmer partners are members of Union Locale de Producteurs de Céréales (ULPC) de Dioila and the Coopérative des semenciers du Mandé (COOPROSEM) (Dalohoun et al. 2011) in direct cooperation with the Association des Organisations Paysannes Professionnelles (AOPP) du Mali. Critical funding came from the German Ministry of Economic Cooperation and Development and McKnight Foundation.

Case study 3: PABRA bean program in eastern Africa

Context. The role and contribution of common beans in Eastern Africa is steadily shifting from a traditional subsistence crop to a market-oriented crop (Buruchara et al. 2011). Yet conventional models of legume breeding and seed delivery in the region remain relatively centralized in national research programs and seed production units (Buruchara et al. 2011; Birachi et al. 2021). As a result of this centralization, farmers throughout the region often lack access to sufficient quantities of quality seed of improved bean varieties (McGuire and Sperling 2016). A key factor contributing to this shortage are the costs associated with complying with country-specific seed policies, laws, regulations, and procedures (Minde and Waithaka 2006).

The policies and regulations that governed bean breeding and seed delivery in each country during the 1990s were designed to meet the needs of public research institutes, and seed quality control systems were built around (and often within) state-owned seed enterprises (ECAPAPA 2002). Each country developed its own regulatory regime for variety testing, approval, and multiplication, and each regime constituted an entry barrier that inhibited new varieties and quality seed from moving between countries, even when cultivation conditions and market opportunities were similar. This led to slow rates of new varieties being released in each country, limited private-sector engagement, the rejection of potentially useful varieties that did not meet national criteria, and significant cost burdens (Krantz 2024).

It was only in the mid-2000s that the two economic communities operating in the region–the Common Market for Eastern and Southern Africa (COMESA), established in 1996, and the East Africa Community (EAC), established in 1999, launched a series of policy initiatives to overcome these barriers. In 2006, the *Agriculture and Rural Development Policy of the EAC (ARD-EAC)* took shape, followed by the Alliance for Commodity Trade in Eastern and Southern Africa (ACTESA) in 2008. Together, these initiatives sought to increase farmer productivity and incomes through trade in strategic agricultural commodities. Since seed itself is an agricultural commodity, COMESA followed up with the introduction of the COMESA Variety Release Systems in 2010, which aimed at harmonizing seed policy across its member countries.

Intervention. PABRA is a consortium of African-owned regional bean networks, international research centers, and their donors, in operation since 1996 (PABRA Pan-Africa Bean Research Alliance 2022). The three regional networks forming PABRA have a combined membership of 31 national bean programs as of 2024. For this study, we review the contribution of PABRA to regional seed system harmonization in eastern Africa, focusing on PABRA's "bean corridor approach" (Birachi et al. 2023). Since PABRA's membership includes public research organizations of EAC and COMESA member-states, there is a clear relationship between PABRA and the regional harmonization efforts.

PABRA's approach is based on the establishment and use of business-centered multi-stakeholder platforms to strengthen bean value chains that allow stakeholders to collectively identify and address bean production, market, and utilization challenges collectively (Gondwe et al. 2024; Buruchara et al. 2011). Within these platforms, PABRA conducts participatory variety evaluation trials to identify farmer- and consumer/market-preferred varieties, expose farmers to improved bean varieties, and accelerate their adoption (Rambiki et al. 2024). Additionally, PABRA uses these platforms to promote knowledge exchanges via learning events that equip farmers with information on productivity-enhancing and climate-resilient agricultural practices (Rubyogo et al. 2019). Tested in Ethiopia early 2000s (Kholová et al. 2024; Rubyogo et al. 2010), the corridor approach was further expanded to other PABRA countries to facilitate the development of public-private partnerships in seed production and distribution (Birachi et al. 2023; Rubyogo et al. 2019). Through its innovation platforms, PABRA contributed to policy dialogues with governments and development partners to improve decentralizing seed quality assurance

systems, trade policies, and financial services for smallholders (Gondwe et al. 2024), although there is no assessment of whether PABRA has actually been successful in changing national policies.

Key actors. The PABRA model comprises partnerships between and among International Center for Tropical Agriculture (CIAT), national agricultural research systems, private-sector actors, and farmers along the region's varied bean value chains. PABRA founding donors are Global Affairs Canada (formerly CIDA), the Swiss Agency for Development and Cooperation and the U.S. Agency for International Development. A set of additional donors also contributed to PABRA over the years and are considered in our analysis.[6]

Table 1 summarizes the context, intervention and key actors of each case study. Despite the three cases being different in institutional contexts, crops of focus, intervention designs and types of actors engaged, similarities emerge from a cross-case reading. First, all case studies focus on historically marginalized communities of smallholder farmers. These communities are overlooked by the formal seed systems, which poorly reflect farmers' preferences and needs in all three cases. Interventions are all focused on enabling reforms in the breeding system and seed governance. PABRA eastern Africa has a unique focus on curriculum advocacy which the two other case studies lack.

Table 1 Summary of context, intervention and key actors for each case study

CIAMO framework	FIPAH Honduras	Sorghum Mali	PABRA Eastern Africa
Context	Marginal hillside smallholders; centralized breeding poorly fits heterogeneous agroecologies	Traditional commons-based seed systems facing tightening certification	Regional policy integration; emerging decentralization; quality-declared seed
Intervention	Farmer research committees; participatory breeding; farmer field schools	Cooperatives linked to public breeding; participatory breeding and seed system development	National/local platforms; quality-declared seed; curriculum advocacy
Actors	Agronomist-led foundation; bean seed committees; national and international breeders; donors	Farmer seed cooperatives; national breeders	National research orgs; cooperatives; private traders; universities; regional bodies

[6] These include the Bill & Melinda Gates Foundation, Australian Centre for International Agricultural Research, the African Development Bank, IDRC, Rockefeller Foundation, Syngenta Foundation for Sustainable Agriculture, AGRA, and KirkHouse Trust.

Cooperatives are key actors in the two African case studies: in the Mali Sorghum case study this refl a specific historical and institutional context, while in PABRA eastern Africa the presence of cooperatives is more project-induced and reflects the market-responsive approach of the program. National research organizations, variety release committees, national breeders and international donors are prominent actors across all case studies.

Results

In this section, we present mechanisms and outcomes of each case study, along the four farmer-first principles of policy change. We summarize trends across case studies in Table 1, at the end of this section.

Case study 1: FIPAH program in Honduras

Inclusive governance. Overall, inclusion of farmers in national agricultural policies has been, at best, consultative in Honduras. The State Policy for the Agri-Food Sector of Honduras (PESAH) 2023–2043 was drafted following a consultation process conducted through almost 50 *Mesas Agrícolas Hondureña* (or Honduran Agricultural Roundtables). From a sampling of consultation notes available online, it appears that some roundtables included farmers' organizations, but it is unclear if they represented smallholder producers. The 2023–2043 State Policy suggests the importance of farmer organization engagement in most government programs, e.g., in the actions around strengthening the work of the Secretariat of Agriculture of Honduras (SAG) in the territories. But overall, the policy text remains vague and lacks directives on how to institutionalize these consultations and engagement further.

In the context of bean production, the 2023–2043 State Policy mentions that in 2018 the *Asociación de Beneficio Mutuo de la Cámara de Frijol* (or Mutual Benefit Association of the Bean Chamber) was created as an association of producers, including approximately 150 members. The policy does not specify further the role of the Bean Chamber. From a journal article online, we read that the group provides a space to coordinate plans and unify criteria with authorities and organizations that contribute to the development of the bean sector in Honduras (Roatan 2018). Similarly, interviews highlight the formal recognition of a National Bean Chain (Seed Change 2018), which FIPAH was vital in helping to establish. "*The Bean Chain takes a leadership role in identifying market and export-oriented opportunities for producers' organizations*", specifies a key informant we interviewed. The respondent continues: "*The [Bean] Chain tries to integrate farmers with intermediaries*

and other actors in the value chain. Many producers in the Chain do not produce large quantities. However, this platform is more focused on the competitiveness of the sector through governance and inclusion of all the actors from the bean chain, to strengthen local and national bean markets". Another key informant noted that, "*Compared to the seed sector, where corporate and institutional interests are strong, the National Bean Chain provided more room for farmers to engage meaningfully in shaping policies*". They amplified this observation by explaining that: *"[…] we have seen incremental changes over time. For example, we have noticed shifts in behavior within certain government organizations. One significant change is a result of the pandemic: that farmers are now allowed to access laboratory analysis for their seeds. Moreover, at the beginning of 2025, communications with government authorities indicated that they are open to integrating the CIALs association into seed production in the formal sector"*.

Responsive Innovation Systems. The incorporation of farmer priorities into plant breeding programs for two important staple crops is primarily due to practitioners' adoption of participatory approaches. Adoption occurred without formal endorsement of these approaches in national policy. One consequence has been that the practice of addressing farmer priorities relies on the conviction and motivation of individuals while their engagement with farmer organizations relies on non-government support, since both participatory breeding and farmer organization depend on funding from international donors. The 2018–2028 National Strategy for the Conservation and Utilization of Plant Genetic Resources of Honduras is the national document that specifies strategies for the development, testing and release of improved varieties in the country. Key actions in the policy include increasing the competitiveness of the bean sector, stabilizing prices, and promoting public-private partnerships. In the latest version of the strategy, CIALs – along with farmers – are mentioned as strategic players at the local level. This comes after years of centralized breeding, where regional testing for bean varieties was conducted by the national program at DICTA (*Dirección de Ciencia y Tecnología Agropecuaria*).

Interviews highlight how this change in the policy is the first step in a long-standing push for engagement, of which FIPAH has been the prime sponsor. As one key informant within the FIPAH project explained: "*A key development is that the main plant breeding program in Honduras now considers participatory approaches as a central method for developing new varieties. This is a major shift, and for me, it's a clear sign that, even without complete recognition from the government, participatory plant breeding is becoming an acknowledged and valuable approach in the field.*"

Relatedly, a representative from one of the donor agencies stated: "*What has always impressed me about FIPAH is how they really put the farmers at the center of everything they do. The people who need to be speaking and who have often less voice they (i.e., FIPAH) put them forward first.*". This sentiment is echoed in remarks from one respondent from the FIPAH project, who explained that "*We now observe farmer field schools integrated in the extension model and in the document [i.e. the 2018–2028 Strategy], and we see statements referring to participation*".

This change in the attitude towards participatory approaches however should not be mistaken for a complete buy-in from the institutions. A respondent engaged in the bean breeding program highlights that "*at the national level it (i.e., participatory plant breeding) has never been accepted*".

Lack of government buy-in is counterbalanced by a strong role of FIPAH's donors, without which a FIPAH respondent specifies that, "*the FIPAH project would not have worked*". For more than 20 years, FIPAH has received funding from Canadian institutions. A representative from one of the donors explained this relationship by stating: "*When I visited as program manager, I was deeply impressed by the grassroots nature of their work, it really stood out. That's what I meant when I spoke about putting farmers first. […] we have always viewed FIPAH as a true partner organization. In fact, FIPAH has significantly influenced [donor name] and its Seeds of Survival program."*

Decentralized Knowledge and Power. In the 2018–2028 National Strategy for the Conservation and Utilization of Plant Genetic Resources of Honduras, it is possible to observe progress, albeit slow, towards the decentralization of Honduras national seed policy. For more than 15 years, FIPAH had a partnership with Zamorano University's bean program which provided foundation seed of participatory plant breeding-generated varieties to CIALs. Consequently, by the time national seed policy began to change in 2020, there was an influential and reputable research institution (i.e., Zamorano University) long familiar with participatory plant breeding. Interviews also show that FIPAH and the CIALs were also able to collect reliable data on the performance of participatory plant breeding varieties.

However, gatekeeping decentralizing seed quality certification procedures by national institutions like DICTA is strong. A recent paper (Gomez et al. 2019) (with the telling title "Pushing back against bureaucracy") details the effort done towards changing DICTA centralized requirements. So far, CIALs-produced seeds must be sold as grain. In 2018, discussions through the National Bean Chain have focused on regional release of bean varieties. Support to accept the idea of regionally approved varieties has been forthcoming from a wide range of institutional actors, but DICTA does not allow the model to be replicated nationally. However, processes are changing faster than codified seed laws. A

respondent within FIPAH provides an example: "*There is another important figure—the director of the Certification Office in the government. This person oversees the entire seed certification process. Last year, he mentioned that while CIALs' seeds are not certified, they can still provide laboratory analysis for farmers who want to sell seeds. […]. They are taking small but significant steps toward incremental changes".*

Transformative education. Agricultural and extension service curricula in national universities of Honduras do not embed formally principles of participatory plant breeding. We find no mention of this topic in national policy. Higher education and the formation of professionals who work in the agricultural sector is not seen by policy makers or educators as a way to promulgate farmer-first approaches, Instead the government envisions farmer field schools as the way to imbed farmer-first principles into the national extension program.

Case study 2: sorghum program in Mali

Inclusive governance. Farmer organizations have become key institutions through which farmers influence policy. The first mechanism of inclusion and empowerment of farmer organizations was to build the capacity of farmer organizations to engage in the full pathway of varietal development and release to seed production, marketing and sale. Farmer organizations have become a one-stop shop for breeders: testing and feedback, with seed multiplication and sale (Christinck et al. 2019). The earlier lack of strict regulations around variety release and seed certification enabled a range of farmer organizations to engage in formal seed production and sale by becoming certified seed producers, especially for sorghum and millet as indigenous crops with farmers' deep cultural knowledge of seed management (Siart 2008). The outcome is that farmer organizations produce roughly 90% of certified seed in Mali, thus being recognized as critical seed system actors (Weltzien et al. 2020). As a respondent observed: "*Yes, the cooperatives produce five times more than the companies. … there's not even a comparison between the two structures in terms of seed production volume.*" Breeders are more comfortable working with farmer organizations than working with private seed companies and engage with farmer organizations in early generation testing. Breeders can also just send seed and get data back, a "*parading shift to considering farmers as partners as opposed to passive beneficiaries*".

Regular meetings are held and they bring together sorghum seed system actors, which "*allow these different actors to come twice a year to discuss these issues of relationship between the local levels and the understanding of these political structures … This platform was really a space for technical debate, i.e. technical debates to influence public policies*". The outcome is that farmer organizations continue to engage with research and development practitioners and are consulted on policy at the local and national level. Farmer organizations – highlights a respondent - "*through their commitment, through their dynamism, they really participate in the action to influence policies at the local level."* Furthermore, farmer organizations have attained such credibility that they get direct funding from donors (e.g. McKnight Foundation and the Agroecology Fund), for example for supporting their own database management. This has shifted power dynamics with donors who state, "*we have already since 2012, given research grants directly to farmers, and they would invite the researchers to support them*".

Responsive Innovation Systems. Farmers' organizations were also able to put in place collaborative management processes. These include: technically trained staff, in the service of a farmer organization act as 'translators' between farmers and researchers; networks of cooperative-based farmer coordinators to ensure effective and routinely communication with researchers and among members, to provide feedback on new sorghum varieties. These processes formed the basis for functioning farmer research networks (Haussmann et al. 2020) despite the level of insecurity in the country (Aina 2025). By linking social and technical innovation processes, farmers organizations influence research, conduct research collectively and jointly with researchers and share the results and learnings (Richardson et al. 2022; Haussmann et al. 2020).

The collaborative learning process became continuous. Farmers, their organizations, researchers and other actors developed tools for data collection, analysis and presentation that allowed continuous adaptation of the breeding program to emerging women's and men's farmer priorities and capacities (Christinck et al. 2019; Isaacs et al. 2024). Farmers can explore new opportunities and researchers understand issues that matter for variety adoption. "*As I speak, research is very much focused on co-creation."* Joint analysis of constraints and opportunities, consideration of gender-differentiated roles and responsibilities in setting breeding trait priorities, data collection in local language, seasonal feedback and planning meetings involving all have become the norm for variety development programs. At the national level, variety releases now also rely on data collected on-farm and include farmers, men and women, among the decision makers. Farmers are represented at research planning and evaluation of IER through research user committees.

The outcome of these changes has been that working with farmers and being accountable to their priorities has become a norm, and the conceptualization of breeding activities is centered on participatory research. As one respondent

succinctly described: "*if they don't want to produce that seed, well, you are not going very far*".

Decentralized Knowledge and Power. In Mali, any agricultural research activity is shaped by differences between agroecological zones (Weltzien et al. 2018). Thus, the targeting of sorghum and millet breeding activities themselves to specific distinct geographies was a necessary mechanism. This essential decentralization of breeding activities was reinforced by organizational and political structures. Farmer organizations are primarily locally based and thus interact with local government services and political actors. Recent political changes also reinforce decentralization, through the creation of new administrative regions, and thus government services and offices in close proximity to farmer organizations' offices. Thus, seed certification has also become more decentralized through local offices of certification and seed labs being built.

Conflict has recently shaped local representation, visibility and voice, and it is much harder for researchers and others to work in the North of the country (Aina 2026). An outcome of strong farmer organizations is that despite conflict they can continue their innovation activities, serving the often drastically changing needs of farmers in their areas, and thus contributing to seed security in their areas. Another outcome of strengthening internal dialogues and learning processes of farmer organizations at the local level has been a focusing on local demands for seed and local value chains, leading to a wide range of diverse varieties of sorghum to address different production priorities and thus contribute to resilience of different types of farmers and household members (Rattunde et al. 2020).

Transformative education. There were two mechanisms through which students were exposed to participatory sorghum and millet breeding programs in the curriculum. First, the two-year, extension-oriented program offered more experiential and on farm learning, and students were more open to working with farmers. The four-year technical program students were less likely to engage in farmer centered work. The second depended on individual change agents, as lecturers who themselves engaged in participatory research brought this to their lessons and took students to local farmer fields for experiential learning. As one respondent noted "[it happens] *if the researcher is convinced, it is an individual thing*". University lecturers observed that changing the curriculum to reflect more participatory research theory and practice was limited by university policy that required layers of approval. There is little evidence around outcomes from the lack of systematic integration of participatory research curriculum, other than farmer organization representatives noting the lack of applied experience in graduating students, and themselves offering internships to students from universities and agro-pastoral schools to fill this gap.

Case study 3: PABRA bean program in eastern Africa

Inclusive governance. A key policy that underpins this case study is the Alliance for Commodity Trade in Eastern and Southern Africa (ACTESA) Strategic Plan 2020–2030, which includes farmers' organization as implementing partners in its implementation organigram. The strategic plan specifies that "*their role will be to mobilize farmers and develop their capacities for effective participation in regional trade policy dialogue and program implementation*" (COMESA/ACTESA 2020, p. 18–19). The Strategy further commits to strengthening and facilitating the development of farmer associations to enhance their capacity to engage in the staple food value chains and improve their participation in national and regional markets. A respondent engaged in the early stages of PABRA development highlights the point that: "*The presence of higher-level farmers' organizations varies tremendously in the region. Since the early days, PABRA worked with cooperatives and grouping of farmers in both seed testing and multiplication. I am not sure if farmers' organizations were engaged at the policy level, but they were engaged as partners in PABRA"*.

Several key informants similarly noted considerable variation in PABRA's engagement with farmers' organizations, and farmers' organizations engagement with policy processes and engagement. For example, one respondent from Ethiopia explained that PABRA engages farmers through one of several mechanisms that the government itself prescribes for projects to support farmer participation in policy and politics. In Zambia, on the other hand, PABRA engages farmers in a manner that is consistent with the national agricultural policy objectives (see MOA Zambia 2011) but primarily focuses its activities on connecting farmers to markets and not on facilitating farmer participation in policy processes.

Responsive Innovation Systems. Although PABRA's efforts to engage farmers did not have a direct influence on farmer participation in policy processes, results strongly suggest that it was effective in incorporating accountability to farmers' priorities. A combination of factors may have contributed to this outcome. First, according to several respondents, was the direct involvement of PABRA leadership and PABRA-affiliated researchers in the *Farmer First* movement. This is because the director of the PABRA bean program and one of PABRA's leading scientists were engaged in the second workshop of the Farmer First Movement, which resulted in the second book (Scoones and Thompson 2009). Second was the introduction to farmer-first principles in the onboarding of new partners and staff in PABRA. As a result, PABRA members were familiar with farmer-first approaches such as participatory varietal selection, farmer-focused engagement in value chain development, and the

importance of linking breeding to farmer livelihood strategies. The use of these approaches inculcated a culture of accountability to farmers' priorities that carried through to the national agricultural research and extension organizations that comprise PABRA's membership.

According to several respondents, a notable shift in PABRA's accountability to farmers' priorities occurred in the 2000s. It was at this time that PABRA began to emphasize accountability to farmers' priorities in the context of the private-sector and market-focused objectives of national agricultural development strategies and investment policies and regional aims to integrate agricultural commodity trade, harmonize seed regulations, and advance a more commercial view of smallholder agriculture. As described by a market researcher who has been part of the PABRA network since 2010, "B*ack then, PABRA was starting to integrate the market-led approach. […] And we also started to build up partnerships for market access … Having the government entities such as extension and even local leaders on the [PABRA] platform was one way to ensure that the policymakers or even policy implementers get firsthand information. So that's one area that we get to bring in the policy as part of that platform whereby you don't have to maybe write up a policy brief and then after that take it to the government units. But rather, the units are part of the process".* PABRA shifted its breeding priorities and activities towards accountability to a broader set of actors in the bean value chain. This shift continued to facilitate farmer access to policymakers and policy processes, while expanding similar access for the private sector, specifically the processors, grains traders, and exporters.

Decentralized Knowledge and Power. The decentralization of research and extension activities figured significantly in the agricultural policies of many PABRA member countries during the past two decades. As a result, PABRA platforms operate nationally and locally. For example, Rwanda's PABRA network is national in its structure, Uganda operates multiple PABRA networks that differ by agroecological zone and bean market destination, while Ethiopia follows a structure that partly aligns with its decentralization of research and extension to regional states. As key respondents from PABRA's leadership indicated, PABRA pursues models that are appropriate to context, consistent with farmer-first principles.

This pattern is also illustrated in PABRA's work on introducing new farmer-facing seed regulations in Uganda. PABRA, alongside Wageningen UR's Integrated Seed Sector Development (ISSD) project, was a significant actor in Uganda's regulatory reform process that allowed farmers' organizations to produce and sell quality-declared bean seed. Seen as a viable alternative to strict and centralized seed certification standards, quality-declared seed is considered a watershed in efforts to decentralize seed regulation and accommodate farmer priorities at a very local level (Bagamba et al. 2023; Mastenbroek et al. 2021).

However, there is a degree of resistance to decentralizing research, extension and seed systems regulations. Respondents interviewed agree that "*the structure of the policy in the area has sort of concentrated and solidified over time, to the advantage of certain players in the seed system. Deregulate and localize would play into the favor of people like the PABRA network*". An informant familiar with seed systems in the region and PABRA underlines: "*What I've seen in other places is that the idea of QDS is quite practical and quite feasible, but not financially or business wise very viable. So, the problem is not the idea, it's the way the business case is developed and the restrictions on the policy*". Informants converge on the idea that the PABRA approach is ultimately constrained by seed policies: their farmers, traders, cooperatives can't necessarily produce and sell seed that is not certified, nor is there much attention to building this capacity.

Transformative education. Several key informants indicated that PABRA has consistently advocated for curricula improvements that focus on farmer-first principles in the agricultural universities and colleges of its member countries. Examples of teaching materials from PABRA affiliates include Persley and Anthony (2017) on The *Business of Plant Breeding: Market-led Approaches to Plant Variety Design in Africa*", as well as a collection of open-access books and training manuals in multiple languages,[7] and other bespoke learning/training event materials that are not online. As a senior manager in Zambia's national agricultural research system said, "*Initially, the University of Zambia didn't have [programs on farmer engagement]. But now with the interaction with the farmers and the extension, for instance, they have introduced a bachelor's degree in agricultural extension specifically to handle the needs of the farmer, mainly the smallholder farmers".* Similarly, a respondent external to but knowledgeable about PABRA explained that: "*Of course, [the work of PABRA] has a huge influence on all of the next generation of breeders, entomologists, pathologists, whatever they will have come into contact with the team or whether they have been inspired by them. […] They're the people in the future who will change a policy*".

Table 2 summarizes findings for each case study along the mechanism and outcome component of the CIAMO framework.

[7] For example, see PABRA's repository of manuals at https://hdl.handle.net/10568/80221.

Table 2 summary of comparative analysis across the three case studies

CIAMO framework	FIPAH Honduras	Sorghum Mali	PABRA Eastern Africa
Mechanisms	Iterative experimentation; trust-building; long-term champions; informal mainstreaming	Cooperative networks reshape seed governance; participatory varietal selection	Seed regulatory reform; multi-stakeholder coordination; curricula
Outcomes	12 bean and 7 maize varieties released; stronger farmer agency; informal institutionalization despite weak codification	Commons-to-formal seed transition; cooperatives improve access; policy lag persists	Regional recognition of quality-declared seed; expanded training; persistent policy resistance

Cross-cases synthesis

Next, we compare patterns across case studies in Table 2, and we map per-case and across cases farmer-first principles institutionalization in Fig. 2. On inclusive governance, in the case of Sorghum Mali and PABRA easter Africa, we find that institutionalization of farmer organizations led them to hold a formal, recognized role in the breeding and seed system. In the case of FIPAH Honduras, policy recognition of farmers' role lags behind and the role of farmer-led institutions remain informal, consultative and donor-relevant. On the principle of responsive innovation system, we observe that integration of farmers' priorities and needs become routine when farmers and/or farmer organization perform seed production or testing functions. This is the case of Sorghum Mali. In the case of FIPAH Honduras and PABRA eastern Africa, we observe that responsiveness hinges predominantly on individual practitioners and motivations. On decentralized knowledge and power, we find that all cases show real decentralization pressure, but national seed certification and variety release bodies continue to gate keep the process from a policy perspective. This holds true in all three case studies. Finally, we observe that the education dimension is the weakest everywhere, advancing meaningfully only in PABRA eastern Africa, and there with a market-led rather than farmer-led emphasis.

The cross-studies synthesis delivers also three themes, which we identified inductively reviewing the evidence collected. These represent reflections on what drives change in institutionalizing farmer-first principles. Implications of these themes are discussed further in section "Discussion".



Fig. 2 Map of the institutionalization of participation along the four farmer-first principle (per-case and across cases). Source: own elaboration, using Affinity Designer

First, the case studies suggest that change is often driven by motivated individuals ("champions"), frequently without institutional buy-in. In every case, seeding change toward farmer-first principles required highly committed individuals: the agronomist-founders and donor partners of FIPAH, the breeders and farmer-organization leaders in Mali, and the farmer-first-oriented leadership of PABRA. This was most visible in the education dimension, which in all three cases advanced (where it advanced at all) through individual lecturers rather than institutional mandate.

Second, change is more likely when farmer-first organizations partner with more influential organizations. The cases show that partnerships with influential institutions—which may not have internalized farmer-first principles but share an interest in getting desired varieties to farmers—were repeatedly decisive. This was clearest in FIPAH's partnership with Zamorano University (FIPAH Honduras case study) and in PABRA's partnerships with national research systems, regional bodies, and private value-chain actors (PABRA eastern Africa case study). Such partnerships extend reach but can also tilt the agenda toward market-led rather than farmer-led ends, as the PABRA eastern Africa case illustrates.

Third, national policy lags behind well-established but uncodified participatory practice. Across all cases, participatory routines became embedded in research and extension organizations well before (and sometimes regardless of whether) they were codified in policy. In FIPAH Honduras and Sorghum Mali, formal seed-system reforms caught up with practices that were already well established; in PABRA eastern Africa, decentralization (notably quality-declared seed in Uganda) advanced furthest where it was formalized, but elsewhere remained constrained by nationally centralized seed policies. A consistent corollary is that grassroots voice rises across all cases even when formal codification lags.

Discussion

Our study explored the extent to which the farmer-first agenda has been incorporated in national agricultural policies in the domains of plant breeding and seed systems. Analyzing evidence along the four farmer-first principles of policy change, we find common threads that we consider representative of the interaction between farmer-first principles and national policy agendas. We discuss those in this section.

Findings on **the farmer-first principles of inclusive governance (the politics of policy) and responsive innovation systems (the politics of demand)** show the precarity of institutionalizing participatory research into policies and practices. On the one hand, we find that change in political winds paired with a lack of buy-in from national policy institutions disrupts the mainstreaming of participatory research models in policy institutions (FIPAH Honduras and Sorghum Mali). On the other hand, across several of our case studies, we observe that when policy buy-in happens, this is frequently too late: participatory practices and routines of national research and extension bodies have already been mainstreamed, even if not codified. The strengthening, capacitation, and mutual learning systems created over time between researchers, farmers' organizations, NGOs, private actors, and donors enables an environment which routinely engages farmers and farmers' organizations. This is particularly evident in the FIPAH Honduras and Sorghum Mali cases, where policy reforms in the formal seed system caught up with unwritten, but well-established practices. We argue that it is through this informal engagement that farmer-first-related projects navigate change and sometimes avoid the introduction of harmful or unsuitable policies.

Paired to this precarity of institutionalizing participatory research is the role of "champions" among researchers, farmers' organizations and donors. In all the case studies analyzed, we find that seeding change towards farmer-first principles required the presence of highly motivated and committed individuals, i.e., champions. This suggests that active and vocal champions applying pressure to policy actors change national agricultural policy and practice, and not the evidence emerging from adapted and adopted varieties. However, the reliance on champions raises questions of sustainability and durability of policy changes, once their commitment ceases. These findings are aligned with (Ashby and Sperling 1995). Finally, we observe that, when farmer organizations shift into the role of seed producer and seed business/traders, they gain visibility and voice in policy arenas. In all the three cases, a shift from on-farm testing for breeding towards seed production, marketing, and sale led to a broader engagement with policy. However, if national policy does not promote growth of farmer organizations hand-in-hand with a formal commitment to a participatory approach, we do not observe much momentum on the expected outcomes. Supply-side participation (promoted by donors and governments) seems to be insufficient, unless the demand-side (farmers) are organizing and have opportunities for agency in some key areas like seed production and marketing.

Findings on **the farmer-first principle of decentralized knowledge and power (the politics of knowledge)** illustrate the role of partnerships between the organizations practicing farmer-first approaches and other more influential organizations (e.g., among private sector actors) which may not have internalized farmer-first principles but have a common interest in the mutually desired outcome of

getting desired varieties to farmers. This was particularly evident in the FIPAH Honduras and PABRA eastern Africa cases. Across all the three cases, decentralizing seed quality assurance systems are desirable for farmers, but numerous national policy barriers exist which keep this power concentrated in the domain of national programs and committees. Finally, in all cases it was apparent that regional testing for varieties, with the aim to develop more adapted varieties suited for different agroecological zones, underpinned the necessity to have more decentralized participatory research and farmers' engagement.

The **integration of farmer-first principles in national agricultural university and college teaching curricula (the politics of andragogy)** was the weakest of the four principles analyzed. This was due to difficulties in changing curriculum (Sorghum Mali case) and path-dependency in teaching skills, with most teachers neither trained nor exposed to participatory approaches (FIPAH Honduras case). The case of PABRA eastern Africa is an exception, with the development of teaching material and courses across the network. However, the attention is on market-led approaches, rather than farmer-led. The lack of integration of participatory principles in curricula, textbooks, courses and teaching staff formative material which we find in the three case studies suggests that farmer-first principles, or participatory methods more broadly, have not yet entered the disciplinary mainstream. On the one hand, the transfer-of-technology theoretical paradigm, where farmers are recipients of innovations, matches a techno-scientific paradigm in plant breeding and agronomy where innovations are developed in a controlled and replicable environment, with a clear experimental design. On the other hand, farmer-first principles require a critical reflection on researchers' and farmers' positionality, power, agency and voice (Colley et al. 2021). This challenges boundaries and incentives of discipline-specific knowledge production.

Finally, from all the three case studies, the importance of long-term (15–20 year) commitments from donors in sustaining farmer-first approaches emerges clearly. In the three case studies analyzed, the role of donors in sponsoring and sustaining protracted farmer-first activities has been essential, but in two case studies out of three (FIPAH Honduras and Sorghum Mali), they are part of what we call, an alternative[8] donor group. Thus, we observe relatively small-scale donors playing a disproportionately large role in sustaining participation and engagement with strategic and sustained investment, unlike the shorter-lived role played by large-scale donors.

[8] We use the term "alternative" to indicate a group of donors which do not appear in the list of CGIAR Trust Fund, but they have a similar global scope for the activities funded. We do not intend the two groups to be in juxtaposition.

The study has limitations. First, we do not address the political economy or social, political and cultural changes (including colonial history) in each country. While we recognize this as a shortcoming in understanding the political *milieu* in which the farmer-first-oriented projects were taking place, a deeper dive into political, historical and cultural context was beyond the scope of this analysis. The meaning and appropriate expression of farmer-first principles are culturally contingent: individualist and collectivist orientations, and differing relationships between farmers and authority, can change what each dimension should look like and how participation is experienced. Our cross-case patterns should therefore be read as transferable hypotheses to be tested against context, not as universal prescriptions. Second, our design does not establish welfare outcomes: we can describe policy and practice change aligned with farmer-first principles, but not whether these changes improved farmer prosperity or food security, or for whom—an important question for future, ideally longitudinal and farmer-centered, research. Finally, we confine our reasoning to plant breeding and seed system policies. We do not address policies of agronomy and agroecology, fields where participatory research and development has a long tradition of experimentation and innovation.

Conclusion and path forward

Thirty-five years ago, the *Farmer First* movement (and its antecedents) presented a new paradigm for agricultural R&D. After nearly four decades of accumulated experience in and evidence from participatory approaches, we investigate if and how national agricultural policies have integrated farmer-first principles. We observe three major thematic findings emerging from this review. First, changes are often driven by highly motivated individuals who operate with a lack of buy-in from policy-relevant institutions. Second, we observe change when organizations practicing farmer-first approaches partner with more influential organizations, which may not have internalized farmer-first principles but have a common interest. Third, we find that national policies are falling behind in codifying well-established participatory practices. We discuss pathways—rather than formulas—toward greater farmer participation in research and extension, reflecting on differences in adherence to farmer-first principles. Based on these findings, we recommend (a) reviewing the role of farmer organizations in seed production, marketing and sale as an effective intervention that enhances farmer participation in policy processes; (b) promoting decentralized seed testing, certification and release schemes to reinforce farmers' involvement, local participation, and a more rapid roll-out of more

adapted—and possibly adopted—varieties; and (c) developing dedicated curricula around participatory research in plant breeding and related fields in national and regional graduate programs, as well as professional training in plant breeding and seed systems.

Finally, we look to the future. With diminishing interest, patience and funding from many donors who have made long-term commitments to farmer-first principles in the past, is farmer-first-oriented policy change on the road to extinction? Or is it the case that this type of policy change is increasingly embedded in processes where donors are less influential? Relatedly, what is the role of governments in organizing, engaging, and learning from farmers in these policy change processes? Our findings suggest that in all cases, there are real benefits to wider use of models, strategies and approaches that promote farmer-first principles in policy spaces. Thirty-five years after *Farmer First*, its underlying principles remain relevant to the future of agricultural R&D and public policy.

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Declarations

Ethics statement The review includes key informant interviews. The interview protocol was shared and reviewed by the Institutional Review Board for Human Participants at Cornell University, which confirmed that this research did not require IRB oversight. This decision tree guideline was followed: https://researchservices.cornell.edu/sites/default/files/2019-05/IRB%20Decision%20Tree.pdf. Informed consent was obtained from all informants included in the study.

Competing interest The authors declare no conflict of interest.

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